

Reciprocal relationships, reverse causality, and temporal ordering: Testing theories with cross-lagged panel models

Charles C. Lanfear

Thiago R. Oliveira

Abstract from panel; we may want to decide on specific illustrative case after we've laid out framework of paper, as we want our case to exhibit every potential issue.

Reciprocal causal relationships are a common feature of criminological theories. For example, police forces tend to use force more often in areas where crime concentrates, while at the same time legal cynicism theory suggests that cumulative exposures to police use-of-force can foster criminal activity. When multiple observations over time are available, cross-lagged panel models are commonly used to estimate these reciprocal effects. This is often done without careful attention to the assumptions that must be satisfied to produce valid estimates, such as correctly specified temporal lags, sufficient inter-temporal variation, and proper accounting for unobserved heterogeneity. Failure to satisfy these assumptions can produce severe issues including spurious associations and parameter estimates that are biased or even reversed in direction. In addition, reciprocal relationships violate causal assumptions based on graphical tools; criminological theories that suggest reciprocal causal relationships usually have an underlying macro-micro mechanism often not accounted for in empirical models. We provide guidance on how to align theory, model specification, and choice of estimator and illustrate this using an empirical example. We use data from Chicago at the census tract level and model the potential reciprocal relationship between police use-of-force and violent crime between 2004 and 2016. We finalise highlighting the importance of criminological theory and careful attention to empirical implications of theoretical premises when investigating reciprocal relationships

Introduction

- Initial hook
 - Tell you how to use a CLPM
 - When do you use a CLPM? [DO this later in detail]

- Use it when you expect reciprocal effects
- Also within and between, but NOT our focus; refer to XXX
- Only dynamic models, not static models – though this is important for static models as well
- Explanation of how what we’re doing is different from recent reviews of CLPM literature, e.g., Zyphur et al. (2020).
 - * Zyphur et al. elaborate a complete generalized cross-lagged panel model that accounts for a wider range of specifications. Here we focus on key problems that appear in applied literature.
- Theory and reciprocity
 - When do we expect reciprocal effects? In context of theory testing
- The role of time
 - What are reciprocal effects *really*?
 - * There are no simultaneous effects
 - * Usually a time problem, either:
 - Ambiguity from theory (i.e., reciprocity is substantive)
 - Competing theory when they are specific (i.e., substantive, empirical adjudication problem)
 - Purely empirical problem due to repeated observations (i.e., reciprocity is a nuisance)
 - Separate theoretical and methodological concerns
 - Strong theories should be specific about timing in causal processes; e.g., alcohol to disinhibition; well-defined, observed biological process (a pint today may give you a headache tomorrow but won’t reduce self-control; zero contemporaneous headache).
- Introduce the CLPM
 - What must a model do to capture these?
 - Use 3 wave model as example

Thiago: I think that sounds great. I’ve always pictured this as going deep in the “what are reciprocal effects *really*?” question. The way I see it is the following. DAGs do not allow for reciprocal effects. Yet, lots of theories loosely suggest reciprocal effects. Some people suggest that this might be a limitation of causal graphs (need to find references for this). But I don’t think it’s a limitation of causal graphs at all, instead I think it’s limitation in how we commonly translate theories into causal graphs. It is possible (even probable?) that all criminological theories that suggest reciprocal relationships are actually correct, but they’re missing a crucial component: time. At close inspection, theories don’t suggest that X causes Y and Y causes X simultaneously; they probably suggest that, in a very simple form, past X causes future Y and that past Y causes future X—and variations of

that. So, one thing we can include here is the role of time in translating theories into DAGs—and, in doing so, more deeply discussing what reciprocal effects really are.

Thiago: I understand this rules out reciprocal simultaneous effects. I'm still not convinced those make sense theoretically (and even statistically, as per Allison).

Two Motivating Examples

This section introduces two empirical examples that motivate our discussion on reciprocal relationships, reverse causality, and temporal ordering. The first is the relationship between occupational attainment and crime involvement, as highlighted by classic studies on developmental and life-course criminology (**sampson1992crime?**). The second is the relationship between perceptions of neighborhood disorder and fear of crime, often studied by neighborhood scholars (**jackson2014experience?**).

- One clearly reverse causality
 - The effect of being employed on offending; this is of policy interest, e.g., jobs programs
 - If you are employed, you are busy and have less chances to offend (outside the workplace anyway) (x->y contemporaneous)
 - Being employed may produce identity change that reduces future offending (x - y lagged)
 - Offending can result in losing your job, e.g., due to being arrested (y->x lagged), but we're not interested in that
- One clearly substantive reciprocity
 - Fear of crime and perceived disorder
 - High fear of crime sensitizes you to local disorder, causing you to perceive more
 - Perceiving disorder may increase your fear of crime
 - BOth are rapidly-occurring updating processes; ambiguous and theoretically interesting which is stronger
- The choice of which to think about it is (almost) entirely theoretical / due to researcher interest
 - e.g., how offending leads to job loss or only interested in how fear changes how you perceive environment

Theoretical and statistical models diagrams

- Suppose you have 3 waves of data

FIGURE: 3-wave theoretical model with two or three panels * CClassic simple model matches the fear / disorder (but VERY FAST) * Model with contemporaneous effects matches employment

Theoretical and statistical models

THIAGO

Good empirical research starts with a strong theoretical foundation. It is from a clearly defined theoretical framework that observable implications can be deduced, and it is those observable implications which can then be contrasted against empirical data (**king1994designing?**; Mayo 2018). Theory-testing empirical research is therefore conditional on a thorough and transparent theoretical model (**nagin2019real?**). Given that most criminological theories presume causal processes, the first step should normally involve deriving a non-recursive model that accurately represents the theorized relationships—e.g., the theoretical model could be represented by a directed acyclic graph (DAG). DAGs can formally represent a nonparametric structural equation model, avoiding functional-form and distributional assumptions (Pearl 2009), and are increasingly used to portray premised causal relationships in criminological theory Oliveira (2024). The DAG should be all-encompassing and include all relevant theorized pathways, including unobserved mechanisms.

Once the theoretical model is specified, it should guide the selection of an appropriate estimator. It is important to distinguish the theoretical model from the statistical model: while the latter should approximate the former as much as possible, the statistical model will unavoidably have to handle imprecision and uncertainty that are bound to exist when dealing with empirical data, such as distributional and parametric assumptions and unexplained covariances due to theoretical ambiguities. When dealing with situations involving potential reciprocal relationships and/or reverse causality, we propose leveraging DAGs for theoretical models and drawing on the structural equation modeling (SEM) framework for estimation (**bollen2013eight?**). A clear theoretical model is a necessary prerequisite for specifying an estimator. While both diagrams will not be exactly the same, we suggest that the premised DAG inform some elements of the SEM. Specifically, uni-directional arrows in the structural model should be non-recursive, as per the theoretical model.¹ For example, a key consideration in this process is the timing of effects: theorized effects can be contemporaneous (i.e., occur quickly) or lagged (i.e., unfold more gradually over time). Yet, while the theoretical model displays a non-parametric representation of theorized causal relationships, the SEM

¹SEMs include a measurement component, which specifies measurement models linking manifest indicators and latent variables, and a structural component, which specifies the relationships between latent variables. For now, we are exclusively referring to the structural component of an SEM. Those include bi-directional arrows, which depict the estimation of a covariance between (the error terms of) two variables, and one-directional arrows, which indicate the estimation of directional association based on regression modeling.

can include bi-directional arrows representing empirically observed covariances between variables not explained by the premised DAG—making the statistical model a recursive model. Simultaneous equations should only be used only as a last resort, given their often tenuous theoretical grounding and the risk of overfitting.

When is a CLPM needed?

A cross-lagged panel model is only needed when reciprocal effects over time are presumed in the theoretical model. When the researcher can theoretically assume no effects of the dependent variable on the independent variable over time, other statistical models will be more appropriate than a cross-lagged panel model (see Imai and Kim 2019). For example, research focused on the effects of a single-event exposure—a common topic in life-course criminology and the role of turning points, such as the long-term effects of school drop-out or marriage on offending ([nguyen2018measurement?](#))—would not benefit from cross-lagged panel models.

Instead, reciprocal effects over time are most often a threat when the exposure is a continuous process that happens repeatedly. There are two situations in which reciprocal effects could emerge from theoretical models, thus highlighting the need of cross-lagged panel models. We name them *reverse causality* and *theoretical reciprocity*.

Reverse causality refers to situations in which researchers are primarily interested in assessing the effects of an exposure on an outcome but are concerned about the possibility that the outcome also influences future exposures. From our empirical example, assessing the effects of occupational attainment on crime involvement is of particular theoretical and policy relevance ([sampson1992crime?](#)); in order to assess that relationship, however, one needs to take into account that offending behavior may also have an effect on future employment, even though examining prospects of job loss following an arrest is not of substantive interest. This is a classic endogeneity problem, in which not controlling for reverse causality introduces **CONFOUNDING (?)** bias. This has been a long concern of econometricians—e.g., the Arellano-Bond estimator, which is essentially a cross-lagged panel model, was developed specifically to control for reverse causality ([moralbenito2019dynamic?](#)). When dealing with reverse causality, researchers have a predefined exposure and an outcome of interest and simply aiming to control away the effects of the outcome on the exposure over time.

What we call theoretical reciprocity, on the other hand, refers to situations in which the directionality of the effects is ambiguous. This could be due to, for instance, competing theories suggesting opposing pathways between two variables (Sturgis et al. 2004). From our empirical example, some researchers argue that perceived disorder leads to increases in fear of crime ([wilson1982broken?](#)) whereas others approach this relationship from the opposite direction, arguing that fear of crime leads local residents to view disorderly signs in the neighborhood as problematic ([jackson2004experience?](#)). This theoretical ambiguity implies that reciprocity is an empirical problem—a question that can be adjudicated by theory testers (Brunton-Smith 2011). When dealing with theoretical reciprocity, researchers often

do not have a predefined exposure and an outcome of interest and instead aim to assess both directions.

In both cases, the theoretical model will highlight the threat of reciprocal relationships. A DAG will not distinguish between reciprocal effects due to theoretical reciprocity or due to reverse causality; this distinction is purely conceptual and should be dictated by the research question. Yet, we sustain that this is an important distinction. While in some cases the empirical solution will be similar, in other situations dealing with theoretical reciprocity or reverse causality may lead to different statistical estimators—we expand on this point below.

CHUCK

Imaginary data

In both cases we have three waves of panel data measured in yearly waves.

- For example 1, we know the updating happens much faster than over course of a year; even if we had daily data, it would still be too fast. COuld do it with lab equipment, but lose real world construct and external validity.
- For example 2, is not as fast, but over course of year, still plenty of time for feedback; if we decreased gaps between waves, we'd begin to rule out reverse causality. Theory can imply how narrow a time period lets you rule out.
- Reality and theory move at one recursive (really continuous) pace, but we get snapshots of data

FIGURE: Time DAG

- Continuous time means you can imagine an infinite number of these “missing waves”; don't be too worried about that, but do read this stuff

SEM diagrams

- SEM path diagrams for estimation
 - The structural part of the SEM should also be a DAG, except in rare circumstances, e.g., pure simultaneous equations; these require very strong identifying assumptions.
 - Work out the diagrams; see if they are the same? Probably are!
- Data observed, and thus you must estimate, from discrete often arbitrarily spaced waves; this produces reverse causality

FIGURE: Time estimator; the classic CLPM

- Explain the classic CLPM as an estimator

- Strict exogeneity

Thiago: Bollen & Pearl 2009 provide a nice discussion about the notational relationship between SEM path diagrams and DAGs. We should also look at Imai & Kim's three papers on causal inference with panel data. They use DAGs and they're so well written. I don't think they discuss reciprocal effects at all (could be wrong), but we could rely on them from a notational point of view in using DAGs to depict temporal processes.

How it can go wrong!

- One para introducing the three key issues

DIVVY UP AFTER

Issue 1: Temporal Order

- Getting lag specification wrong is a big issue
- This assumes true or approximately true lag specification
- Unusual form of bias as documented by Vaisey & Miles that can flip signs:
 - If true model is $y = \beta x_t + \alpha_i + e_{it}$ and you fit $y = \beta^* x_{t-t} + \alpha_i + e_{it}$, $E(\beta^*) = -0.5\beta$, i.e., as Allison (2022) says “a positive contemporaneous effect of x gets transformed into an artifactually negative lagged effect”.
- Probably the most underappreciated issue as it may produce misleading results that reject or overturn theories
 - Papers that find negative effects when theory suggests positive ones may be the result of this bias.
- Key takeaway: Don't just be worried about your research but also be skeptical about interpreting other research with unexpected sign flips!

Solutions

- Theoretical ideal: Get it right
- Vaisey & Miles: Do both lagged and contemporaneous
 - If no contemporaneous effect, only lagged, you're probably fine. If contemporaneous shows up, you can't definitively determine direction of contemporaneous effect.

- Allison: Partial correlation model with lagged effect and contemporaneous residual covariance
 - This is least biased approach if parameter of interest is βx_{t-1} , but cannot estimate βx_t
 - This still does not adjudicate between direction of contemporaneous effects; if effect is entirely contemporaneous, all relationship will be in residual covariance.
- Also good place to bring in predetermined regressors which are part of Allison approach and Arellano-Bond

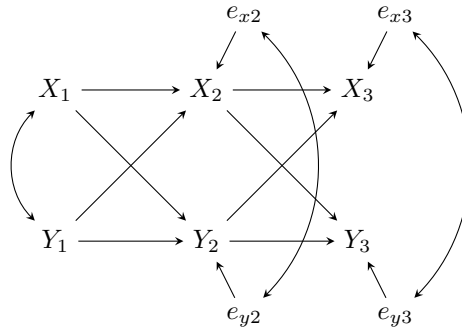


Figure 1: Allison (2022) recommendation

Issue 2: Unobserved heterogeneity

- Probably the most commonly encountered issue; but also pretty much solved!
- Most common concern is stable traits
 - While really same problem, we treat time-varying and time-stable confounders as different problems in longitudinal models
 - Autoregressive term alone does not solve problem
 - Can't just toss in fixed effects due to Nickell bias from endogenous lag (shrinks in proportion to T); Arellano-Bond (and thus Allison) is removing Nickell bias
- Including time-stable covariates in FE models
- Might also raise treatment heterogeneity here as different but related issue not as easily addressed in CLPMs as in TWFE
 - DEvelopmental theories often specify age-specific (life stage) treatment heterogeneity
 - Freeing the parameter by wave allows for time heterogeneity but also effects to differ because wave spacing is different

- * One reason for treatment heterogeneity among individuals is differing measurement times within waves
- Exception is time-specific treatment heterogeneity which is easily permitted by allowing wave-specific parameters as in Allison’s model
- A potentially useful application for wave-specific parameters is when you have unevenly spaced data; I think latent growth models would get at this too but I’m not very familiar with them.
- Predetermined regressors (weakly exogenous, sequentially exogenous)

Solutions

- Largely solved problems!
- RI-CLPM and related approaches
 - These are correlated stable trait models; I’m not very familiar
 - Seems to address time stable unit heterogeneity
- Allison’s FE dynamic panel model
 - Can handle both under same assumptions and problems
 - Allison’s ML-SEM approach is related to this Mundlak model; augments it with predetermined regressors.

The other alternative to the traditional cross-lagged panel model that actually controls for unobserved stable heterogeneity is Hamaker, Kuiper, and Grasman (2015)’s Random Intercept Cross-Lagged Panel Model (see also Mulder and Hamaker 2021).

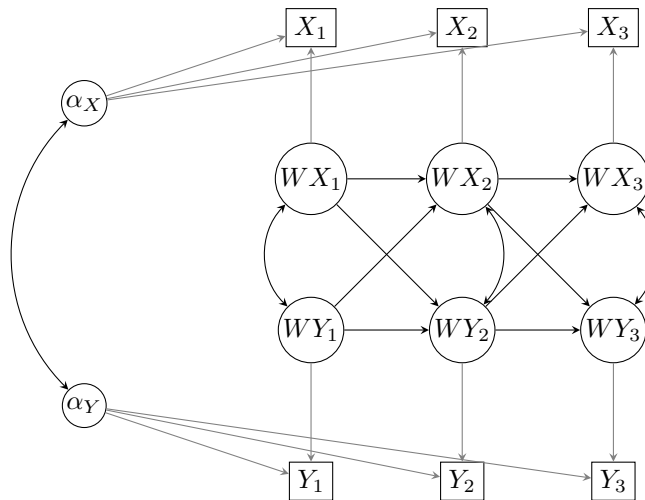


Figure 2: Random Intercept Cross-Lagged Panel Model (Hamaker et al., 2015)

Issue 3: Inter-temporal variation

- Maybe show some illustrative math here to show that $Var(Y_2|Y_1) \rightarrow 0$ as $\rho(Y_1, Y_2) \rightarrow 1$ approaches 1; i.e., you rapidly run out of anything to explain.
 - This is essentially a multicollinearity problem: Unbiased and consistent but high variance in estimates makes any particular estimate suspect.
 - From time series lit: Stronger your autoregressive term is, the less there is to explain; rapidly approaches a time-stable
- Rarely appreciated outside time series literature but common when waves are narrowly spaced
- With random measurement error, the error becomes proportionally larger as time periods get narrower
 - May mean you can address in part with measurement models
 - Note the endogenous variables are regressors, so measurement error biases parameter estimates as well
- Psychological constructs can be very stable over time! Life course trajectories are real!
 - People even get more consistent over time

Solutions

- This is first and foremost a data problem, so you want to attack it during design of data collection if possible
 - Sample to maximize change
 - Look for shocks
- Can put time stable covariates using hybrid approach, but be aware, you're now looking at between-individual effects
- If you can't collect new data, look at different aggregations or wave skips
 - Small change at a large unit may mask larger changes in small units
 - Should prob only consider skipping waves if it makes sense theoretically and/or you're facing serious empirical underidentification problems.
- IF increasing error is what you're worried about, can potentially use measurement models
 - Should be doing this anyway because your outcomes are regressors, so measurement error attenuates estimates
- Give up and go get a pint

General Advice

- Separate theory problems from estimation problems
 - Theory comes first
 - Then a clear model
 - Then estimator
- Default to robust estimators
 - Allison’s FE model should prob be the default as it is most robust
- Also need to think about all these issues not specific to CLPM or reciprocal effects
 - Sequential ignorability
 - Dynamic effects
 - * Other related strategies, e.g., for single outcomes with continuous time-varying confounded exposures

Conclusion

Thiago: Sounds great!

- Brunton-Smith, Ian. 2011. “Untangling the Relationship Between Fear of Crime and Perceptions of Disorder: Evidence from a Longitudinal Study of Young People in England and Wales.” *The British Journal of Criminology* 51 (6): 885–99.
- Hamaker, Ellen L, Rebecca M Kuiper, and Raoul PPP Grasman. 2015. “A Critique of the Cross-Lagged Panel Model.” *Psychological Methods* 20 (1): 102.
- Imai, Kosuke, and In Song Kim. 2019. “When Should We Use Unit Fixed Effects Regression Models for Causal Inference with Longitudinal Data?” *American Journal of Political Science* 63 (2): 467–90.
- Mayo, Deborah G. 2018. *Statistical Inference as Severe Testing: How to Get Beyond the Statistics Wars*. Cambridge University Press.
- Mulder, Jeroen D, and Ellen L Hamaker. 2021. “Three Extensions of the Random Intercept Cross-Lagged Panel Model.” *Structural Equation Modeling: A Multidisciplinary Journal* 28 (4): 638–48.
- Oliveira, Thiago R. 2024. “Aggressive Policing and Undermined Legitimacy: Assessing the Impact of Police Stops at Gunpoint on Perceptions of Police in são Paulo, Brazil.” *Journal of Experimental Criminology* 20 (1): 83–121.
- Pearl, Judea. 2009. *Causality*. Cambridge University Press.
- Sturgis, Patrick, Peter Smith, Ann Berrington, and Yiongjean Hu. 2004. “Using Repeated Measures Data to Analyse Reciprocal Effects: The Case of Economic Perceptions and Economic Values.” *ESRC Methods Festival*.